

Minimal GitHub Guide (HTTPS Version)

Introduction

GitHub is a cloud-based platform that uses **Git** for version control, allowing you to store, track, and manage projects efficiently. Whether you're creating Linux scripts, automation tools, web applications, Docker containers, configuration files, or technical documentation, GitHub helps maintain a complete history of your work and provides a reliable backup of your projects online.

Using GitHub allows you to:

- Track changes to your files over time
- Maintain version history
- Collaborate with other users
- Back up projects safely
- Revert mistakes when needed
- Access your projects from anywhere

This guide focuses on the essential GitHub workflow using **HTTPS authentication**, making it simple for beginners to connect to GitHub without configuring SSH keys.

By completing these modules, you'll have a reliable GitHub setup that you can use for any future project.

You will learn how to:

- Configure Git on a new Linux computer
- Set up a global Git ignore file
- Authenticate with GitHub using HTTPS and a Personal Access Token (PAT)
- Copy an existing repository and make it your own
- Push and pull changes safely
- Manage repositories using a simple daily workflow

Prerequisites

Before starting, ensure you have:

- A GitHub account
- Internet access
- Administrator or sudo access
- Git installed
- A GitHub Personal Access Token (PAT)

Installation

Linux (Ubuntu/Debian)

Update package repositories:

```
sudo apt update
```

Install Git:

```
sudo apt install git -y
```

Verify installation:

```
git --version
```

Module 1: Initial Git Setup (One-Time)

Configure your Git identity.

```
git config --global user.name "Eugene Horfilla"  
git config --global user.email "mreugene2005@gmail.com"
```

Verify your configuration:

```
git config --global --list
```

Expected output:

```
user.name=Eugene Horfilla  
user.email=mreugene2005@gmail.com
```

Module 2: GitHub HTTPS Setup (One-Time)

GitHub no longer accepts account passwords for Git operations. HTTPS authentication uses a Personal Access Token (PAT).

Step 1: Configure Credential Storage

```
git config --global credential.helper store
```

Verify:

```
git config --global --list
```

Expected output includes:

```
credential.helper=store
```

Step 2: Create a Personal Access Token (PAT)

Navigate to:

```
GitHub
├── Settings
│   ├── Developer Settings
│       ├── Personal Access Tokens
│           ├── Tokens (Classic)
│               └── Generate New Token
```

Recommended permissions:

```
repo
workflow (optional)
```

Generate the token and save it securely.

Step 3: Test Authentication

Clone a repository:

```
git clone https://github.com/YOUR_USERNAME/YOUR_REPOSITORY.git
```

When prompted:

Username: YOUR_GITHUB_USERNAME

Password: <PASTE_YOUR_PERSONAL_ACCESS_TOKEN>

Git will save your credentials for future use.

Module 3: Copy an Existing Repository and Make It Your Own

Use this process when you want to take an existing GitHub project and create your own independent version.

Step 1: Clone the Original Repository

```
git clone https://github.com/OTHER_USER/project.git
```

Example:

```
git clone https://github.com/dikoalam18/echo_linux101.git
```

Enter the directory:

```
cd echo_linux101
```

Step 2: Remove Original Git History

```
rm -rf .git
```

This removes the connection to the original repository.

Step 3: Initialize a New Repository

```
git init
```

Step 4: Stage Files

```
git add .
```

Step 5: Create Initial Commit

```
git commit -m "Initial commit"
```

Step 6: Create a New GitHub Repository

Create a new empty repository in GitHub.

Example:

```
linux101-copy
```

Important:

- Do NOT create a README
- Do NOT create a .gitignore
- Do NOT create a license

Leave the repository empty.

Step 7: Connect Your Repository

```
git remote add origin https://github.com/YOUR_USERNAME/linux101-copy.git
```

Example:

```
git remote add origin https://github.com/dikoalam18/linux101-copy.git
```

Verify:

```
git remote -v
```

Expected:

```
origin https://github.com/dikoalam18/linux101-copy.git (fetch)
origin https://github.com/dikoalam18/linux101-copy.git (push)
```

Step 8: Rename Branch to Main

```
git branch -M main
```

Step 9: Push to GitHub

```
git push -u origin main
```

Expected:

```
Branch 'main' set up to track 'origin/main'
```

Quick Reference

New Computer Setup

```
git config --global user.name "Eugene Horfilla"  
git config --global user.email "mreugene2005@gmail.com"  
touch ~/.gitignore_global  
git config --global core.excludesfile ~/.gitignore_global  
git config --global credential.helper store
```

Create a GitHub Personal Access Token (PAT).

```
git clone https://github.com/OTHER_USER/project.git  
cd project  
rm -rf .git  
git init  
git add .  
git commit -m "Initial commit"  
git remote add origin https://github.com/YOUR_USERNAME/project.git  
git branch -M main  
git push -u origin main
```

Save Your Work

```
git add .  
git commit -m "Updated project"  
git push
```